L2-Basics of Programming

Special class

Love Babbar • Apr 3, 2025

L2 - Basics of Programming

- Variables and Operators
- Write your 1st Program

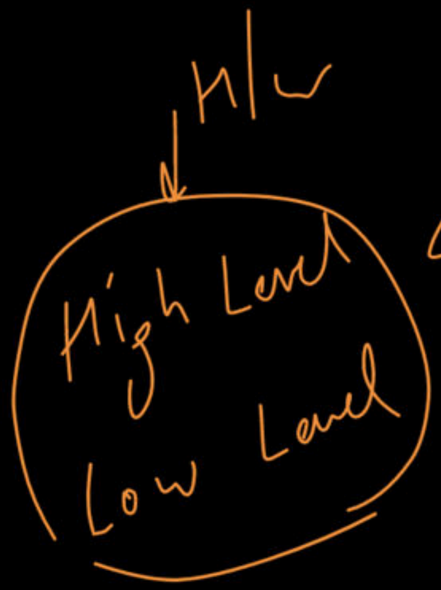
—> by Codehelp

Programming Language ?

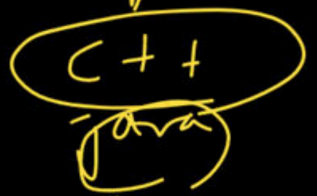
What ?

Why ?

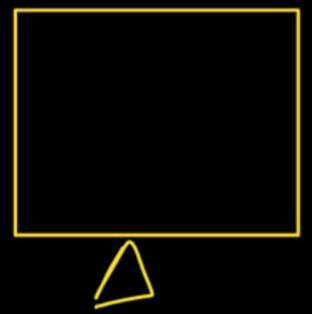
Standard



Programming language

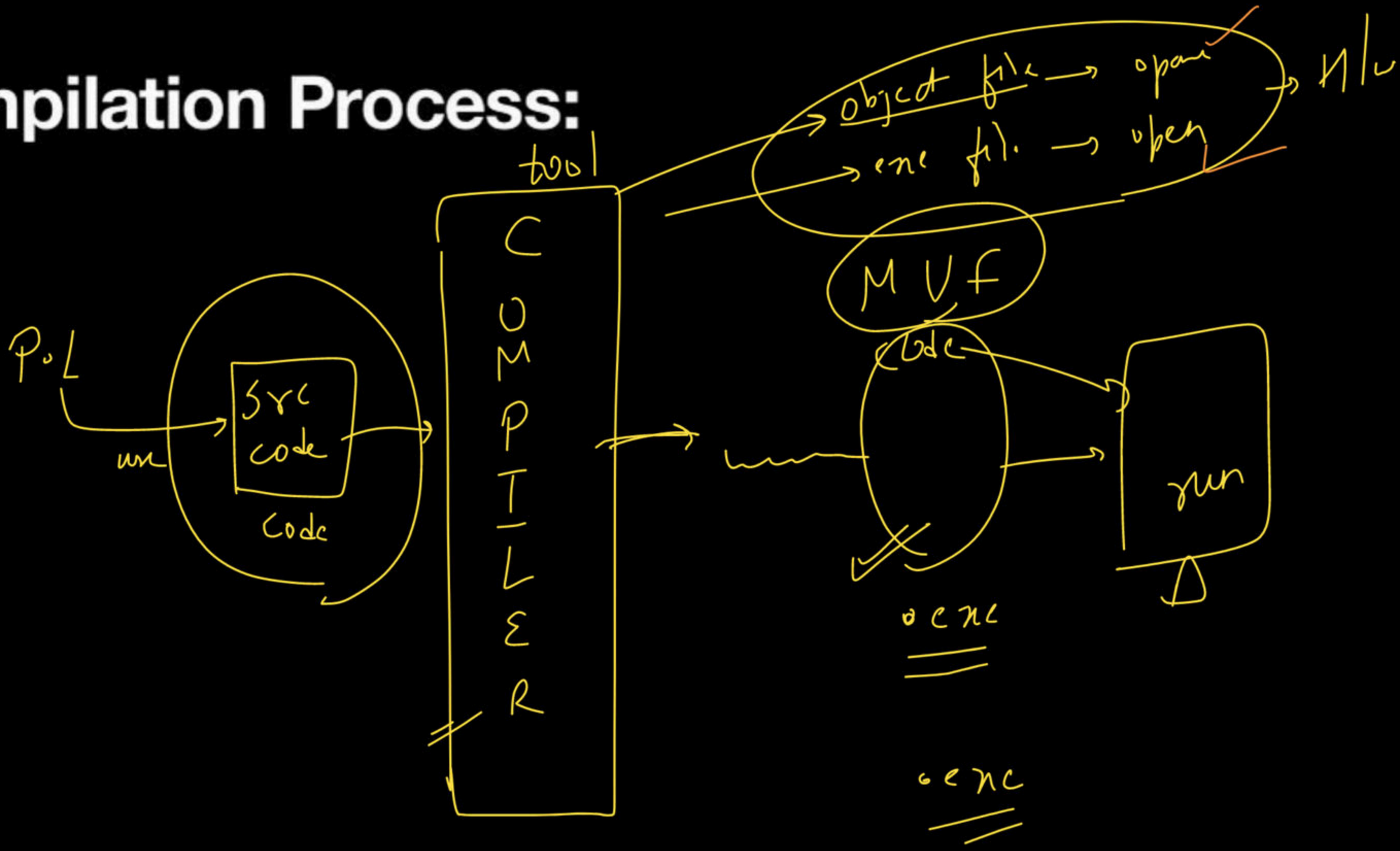


Python

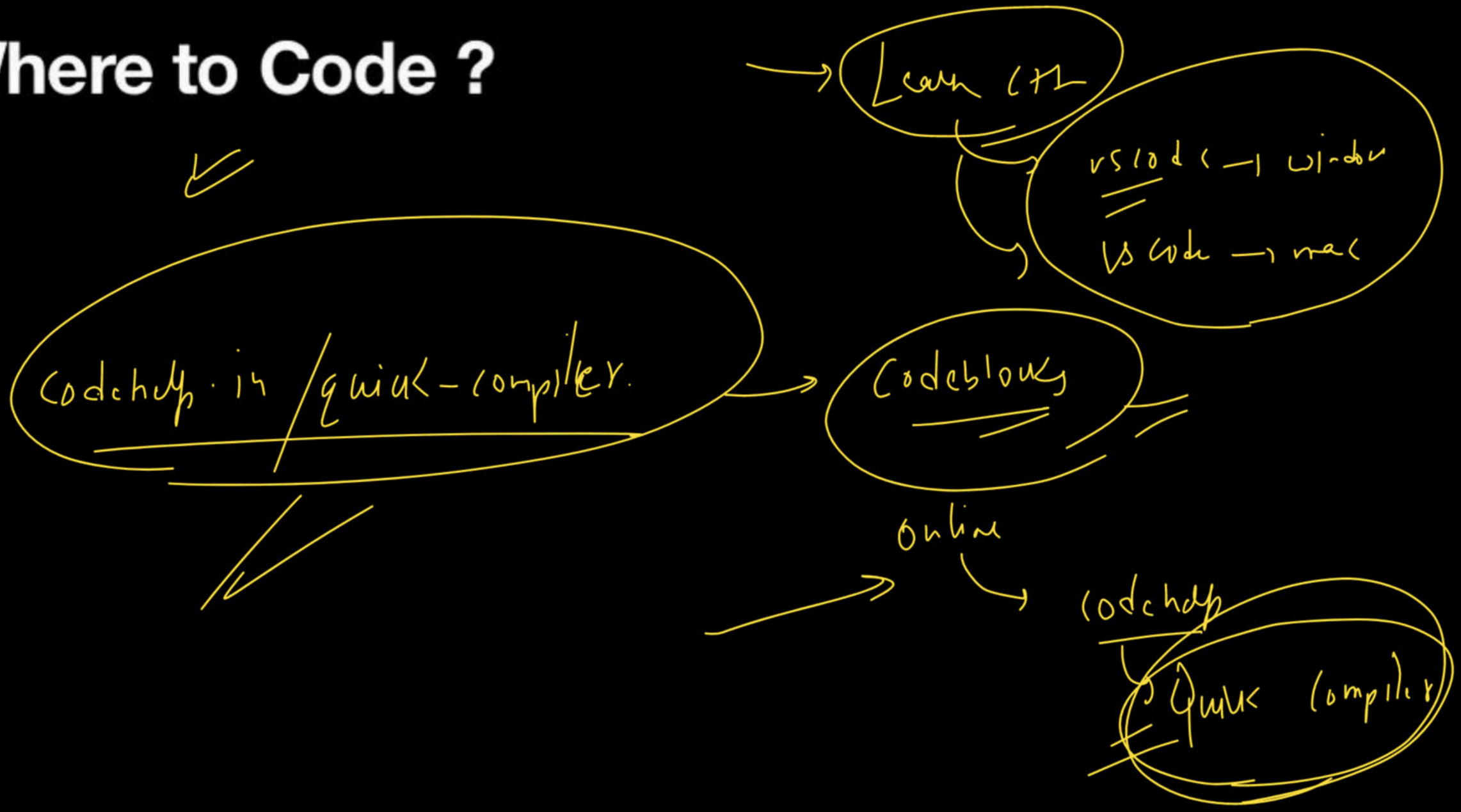


task

Compilation Process:



Where to Code ?



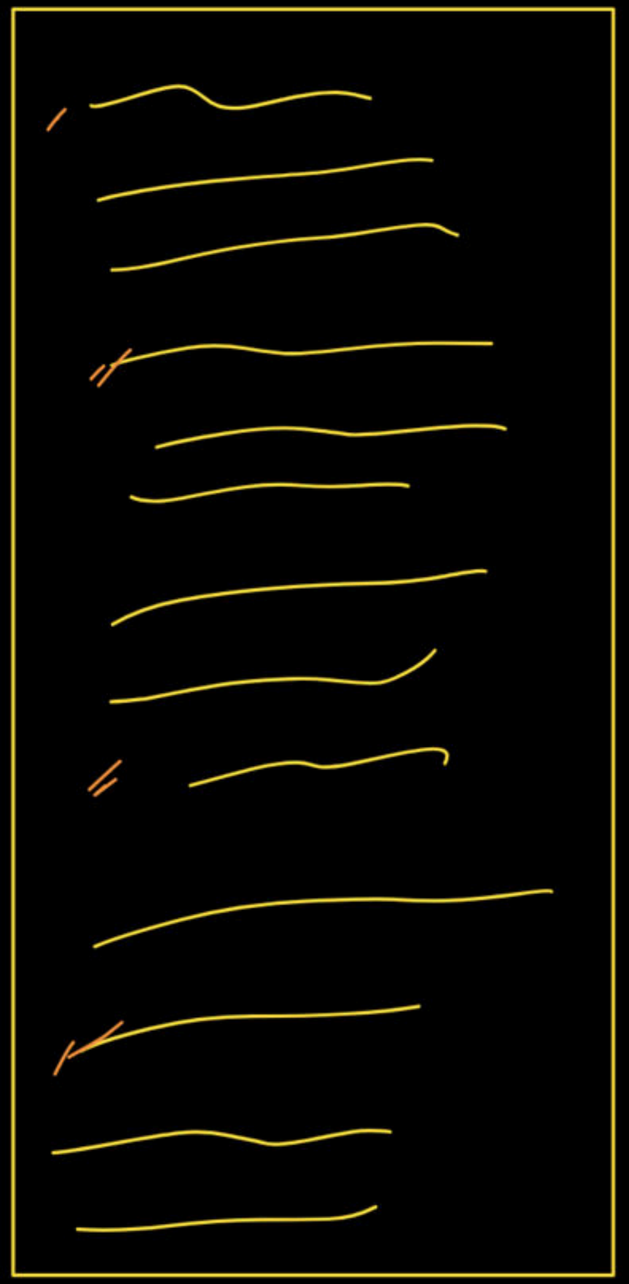
Your 1st Code:

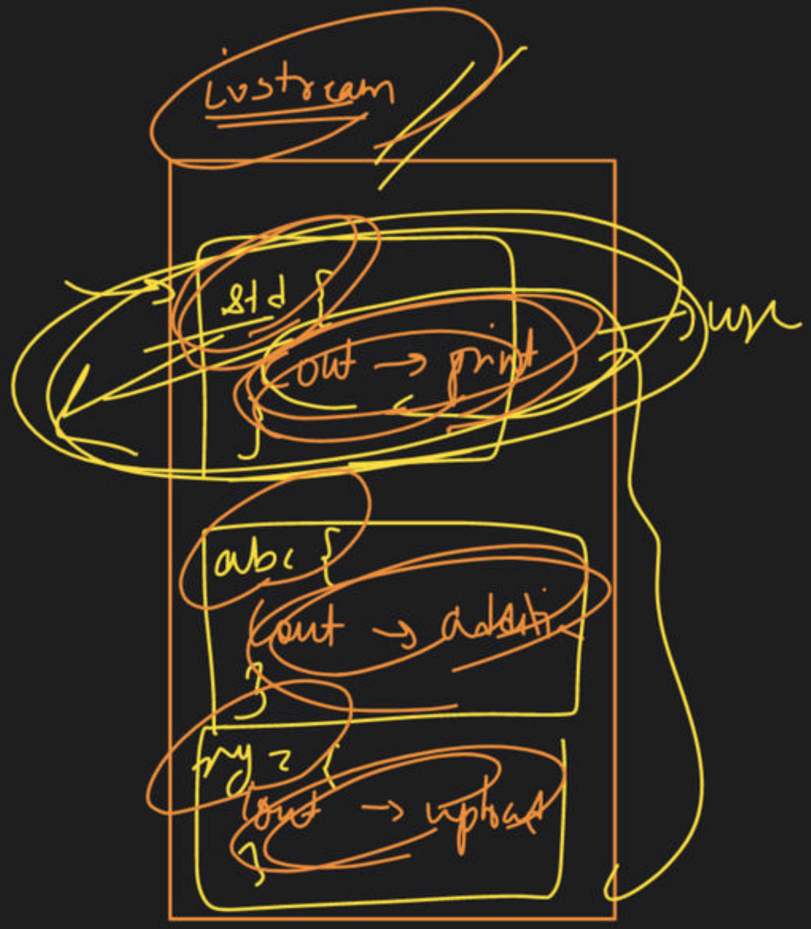
C++

↳ starting point

int main

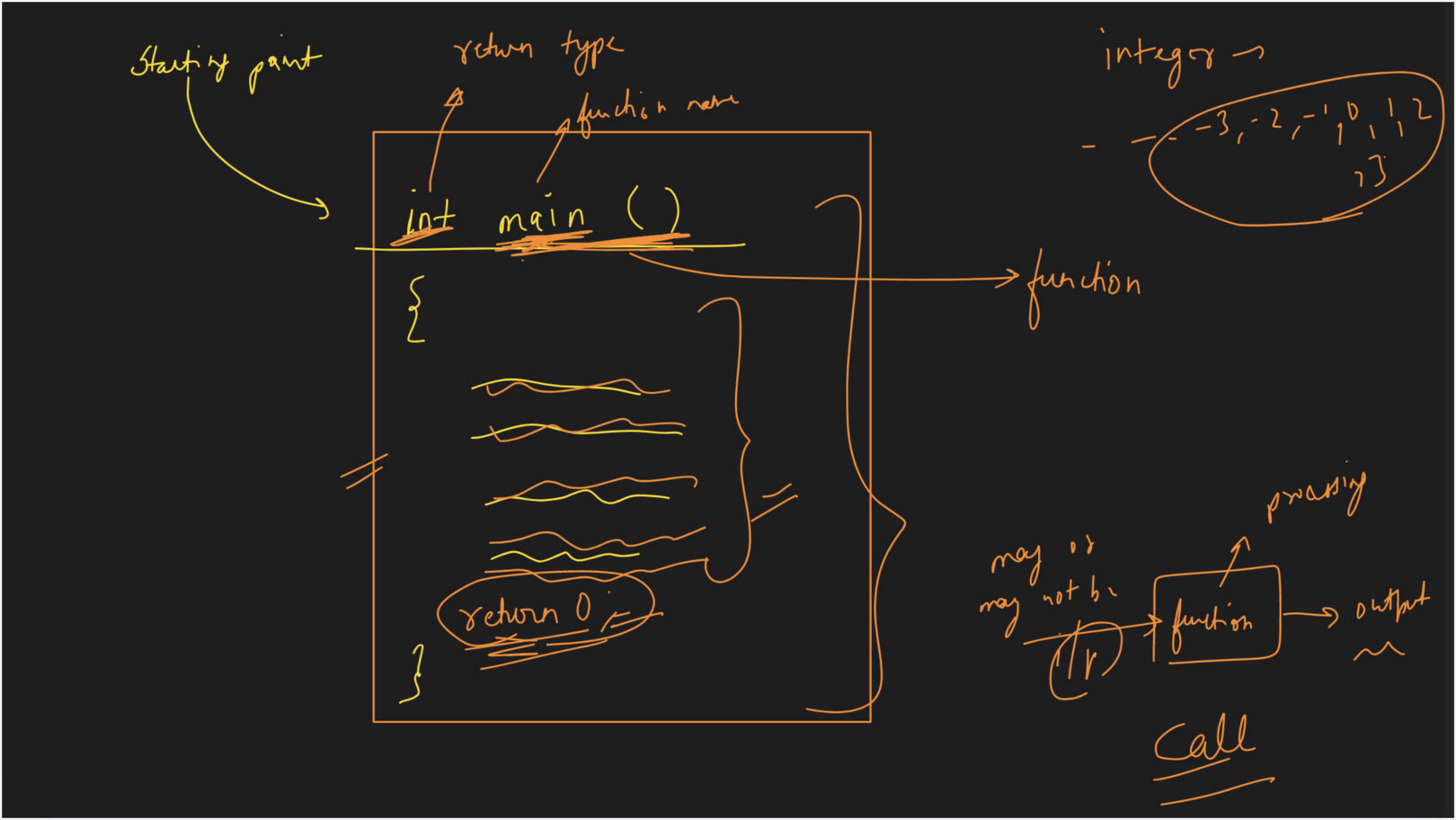
Starting point

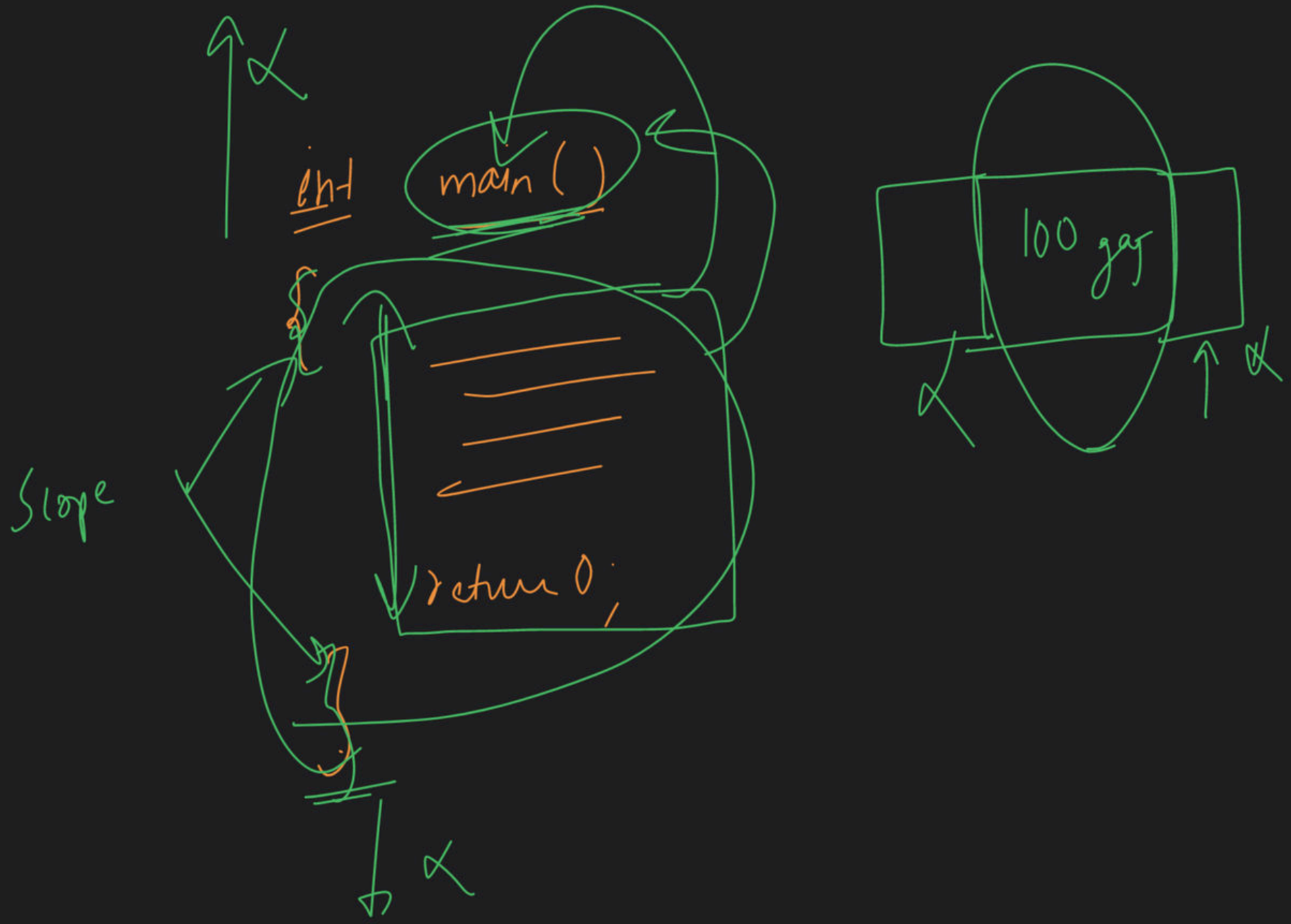


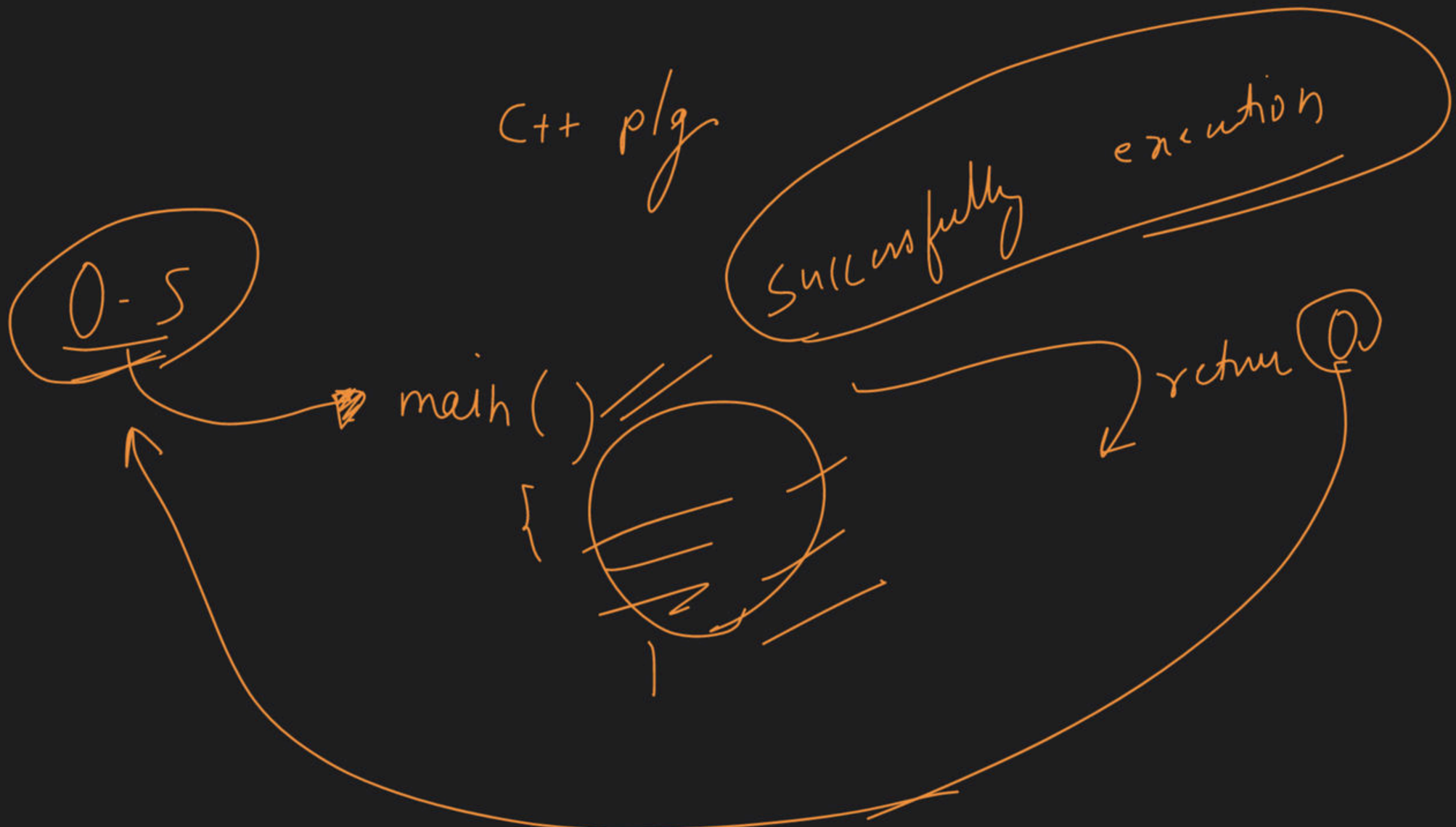


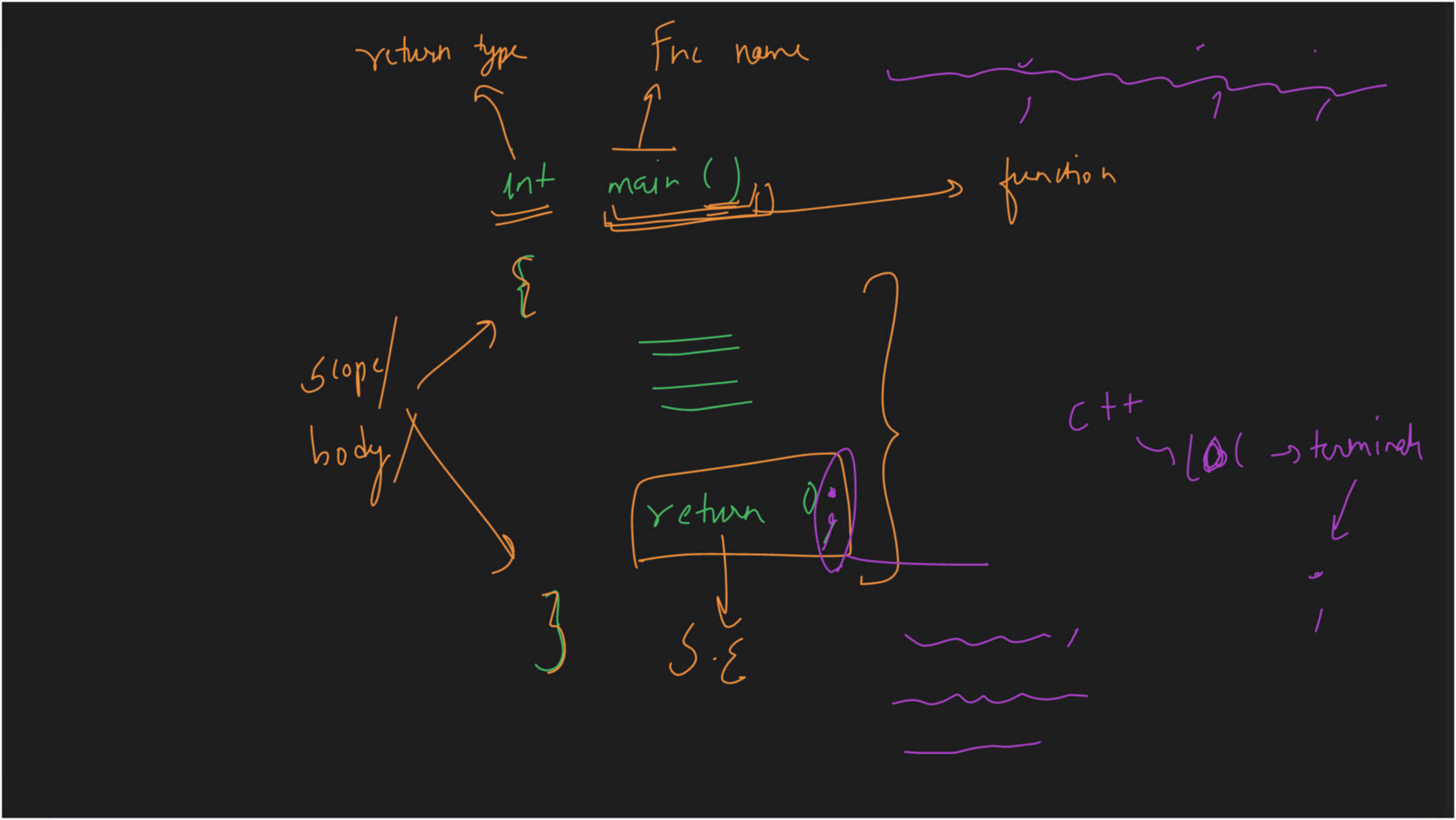

```
#include <iostream>
using namespace std;

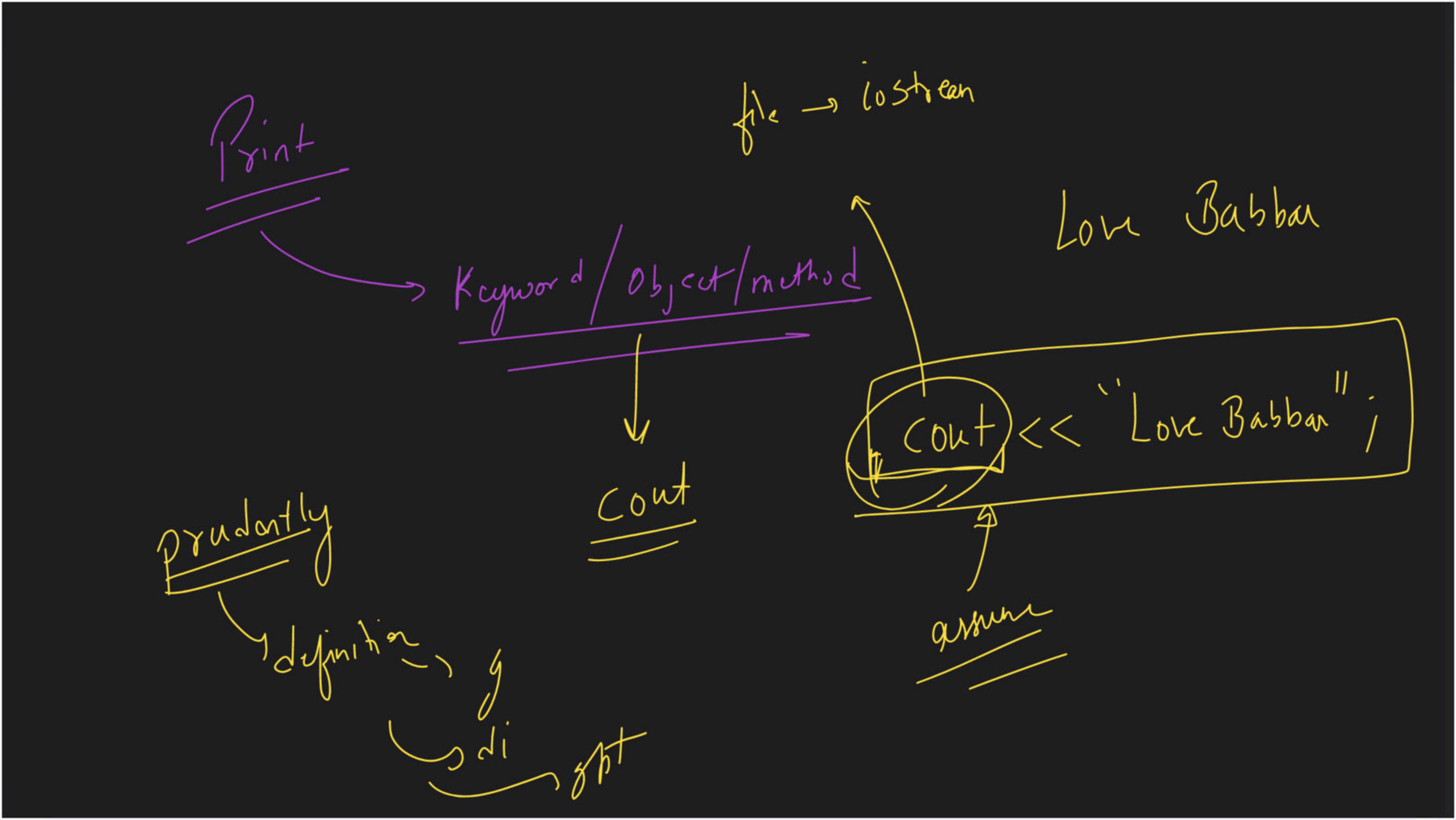
int main()
{
    cout << "Love";
    return 0;
}
```

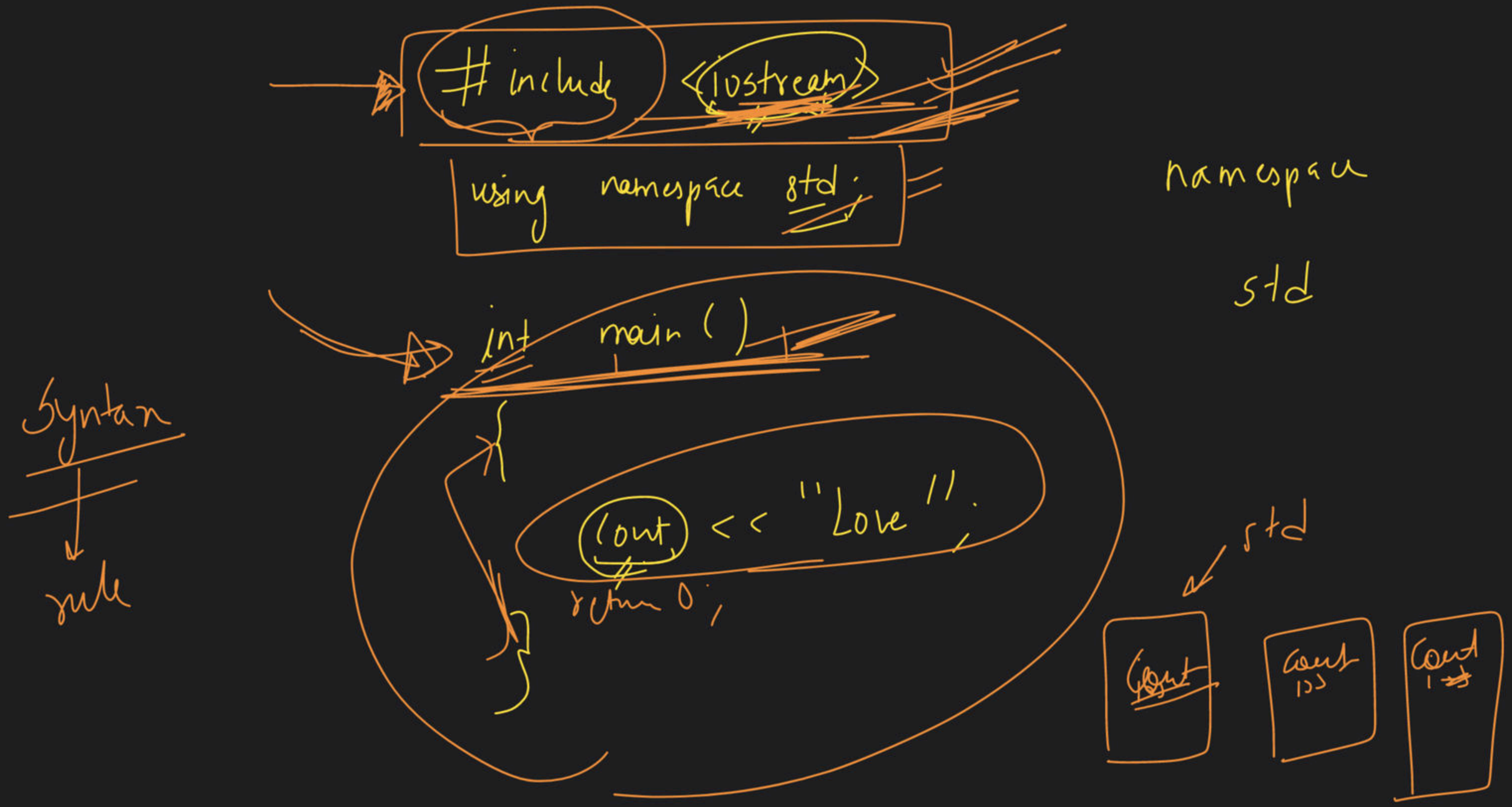

Starting point
return type
function name
integers →
-3, -2, -1, 0, 1, 2, 3
int main ()
{
return 0;
}
function
passing
may or may not be
function
output
Call

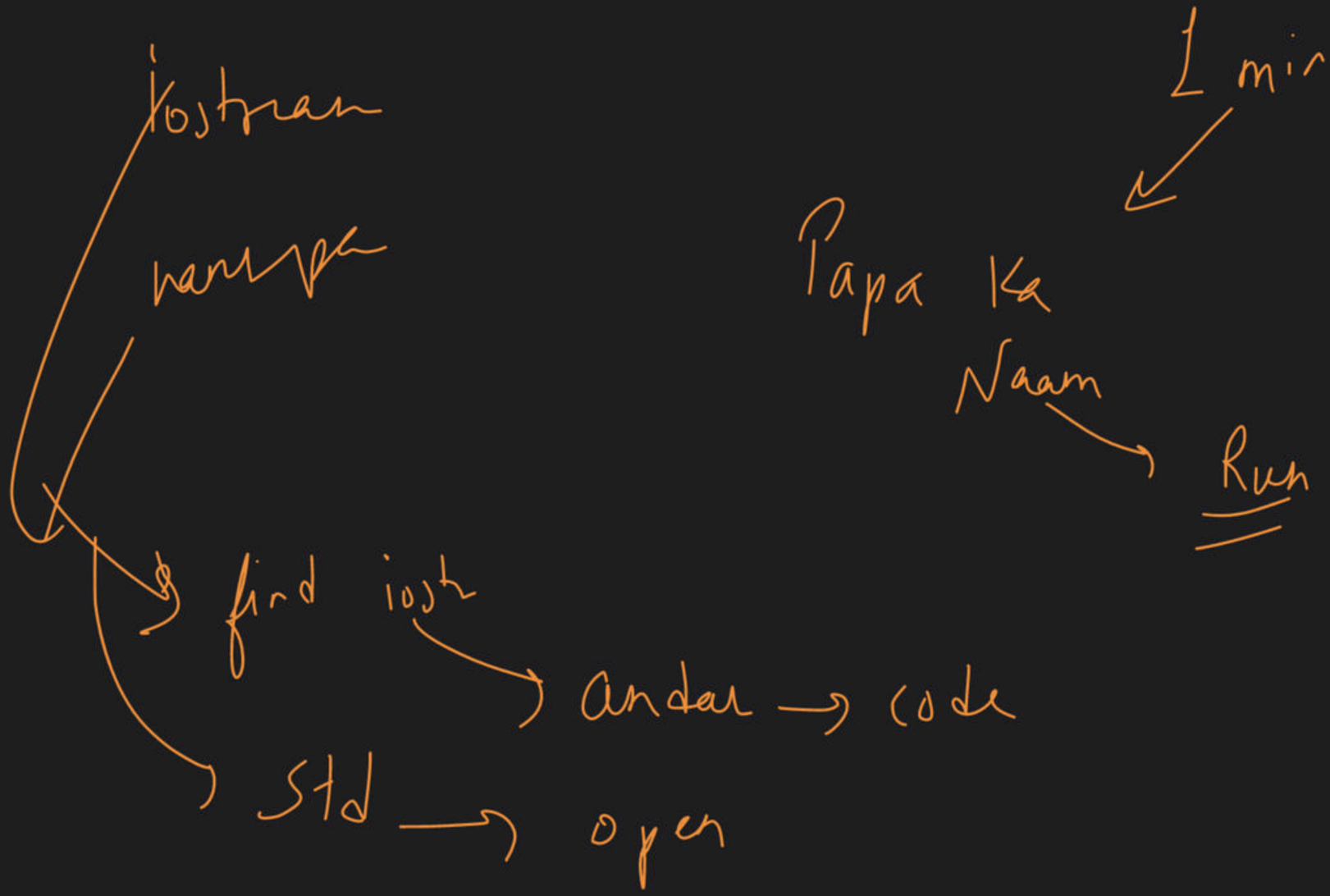

int
main ()
{
return 0;
}
Slope
100 gaj

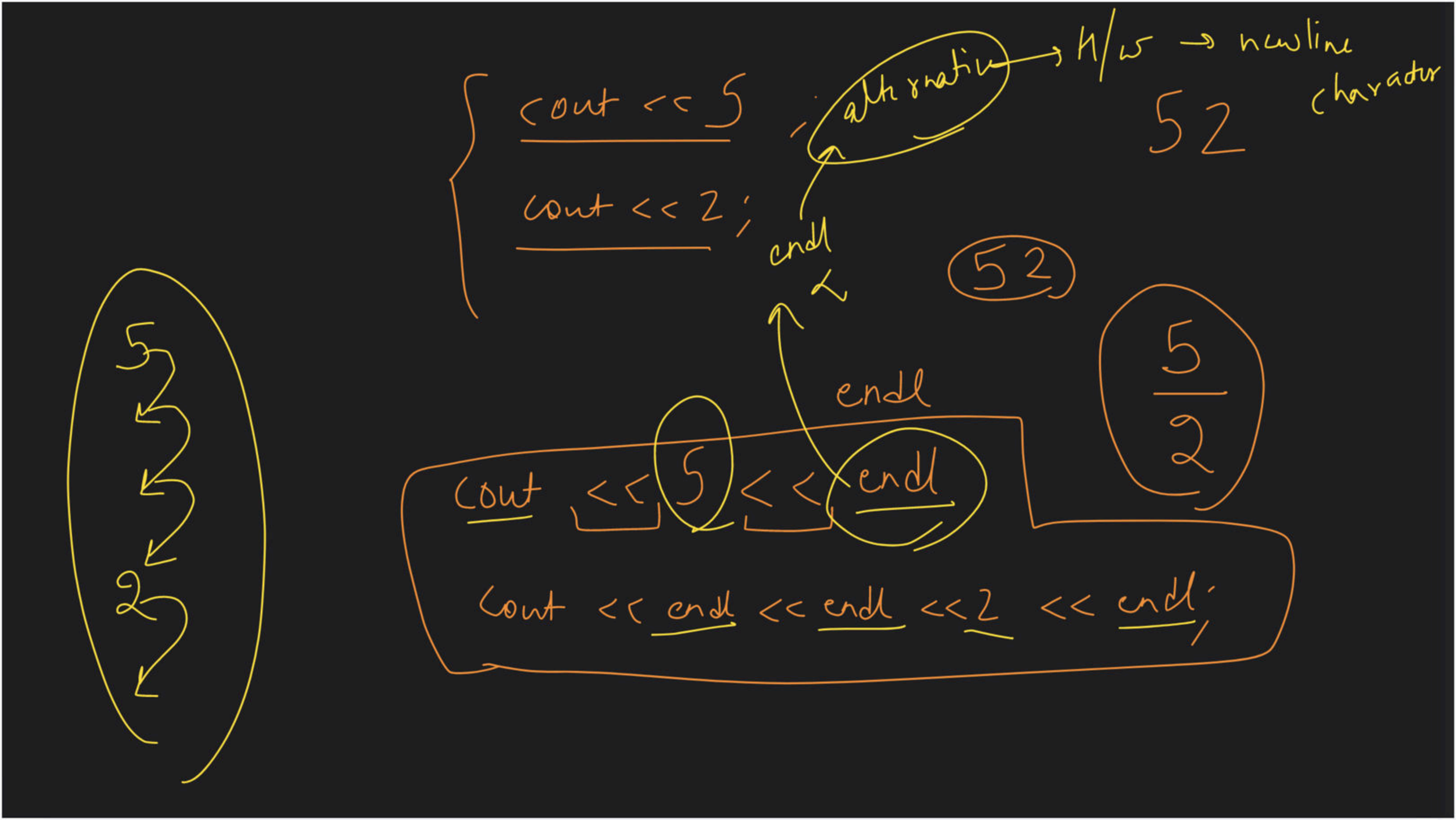

C++ p/g
O.S
main()
{
}
successfully execution
return 0


return type
Fnc name
int
main ()
function
{
scope/
body
return 0;
}
S.E
c++
LOL -> terminate

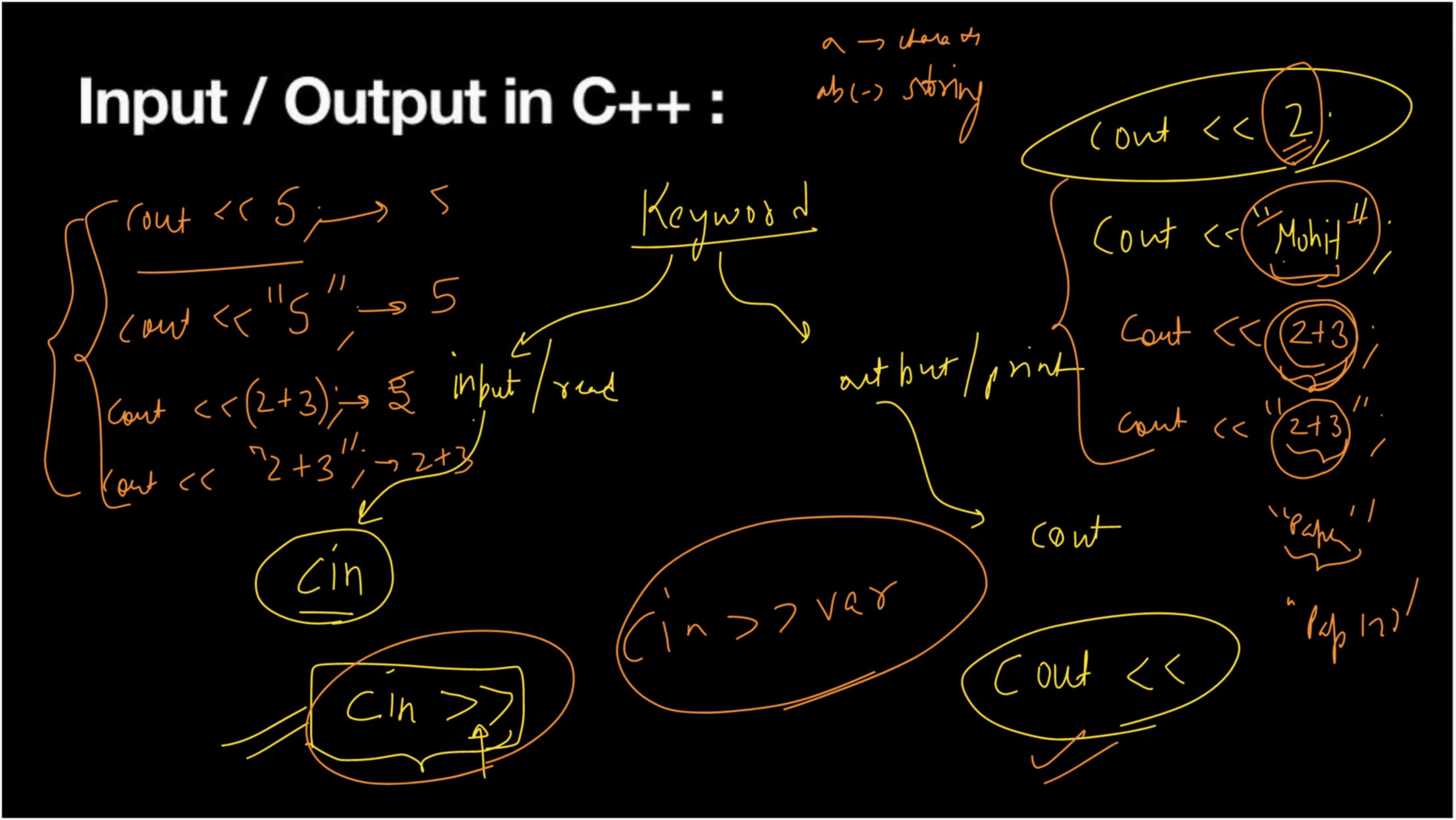

Print
file → iostream
Love Babbar
Keyword / Object / method
cout
cout << "Love Babbar" ;
Prudently
definition
gpt
assume

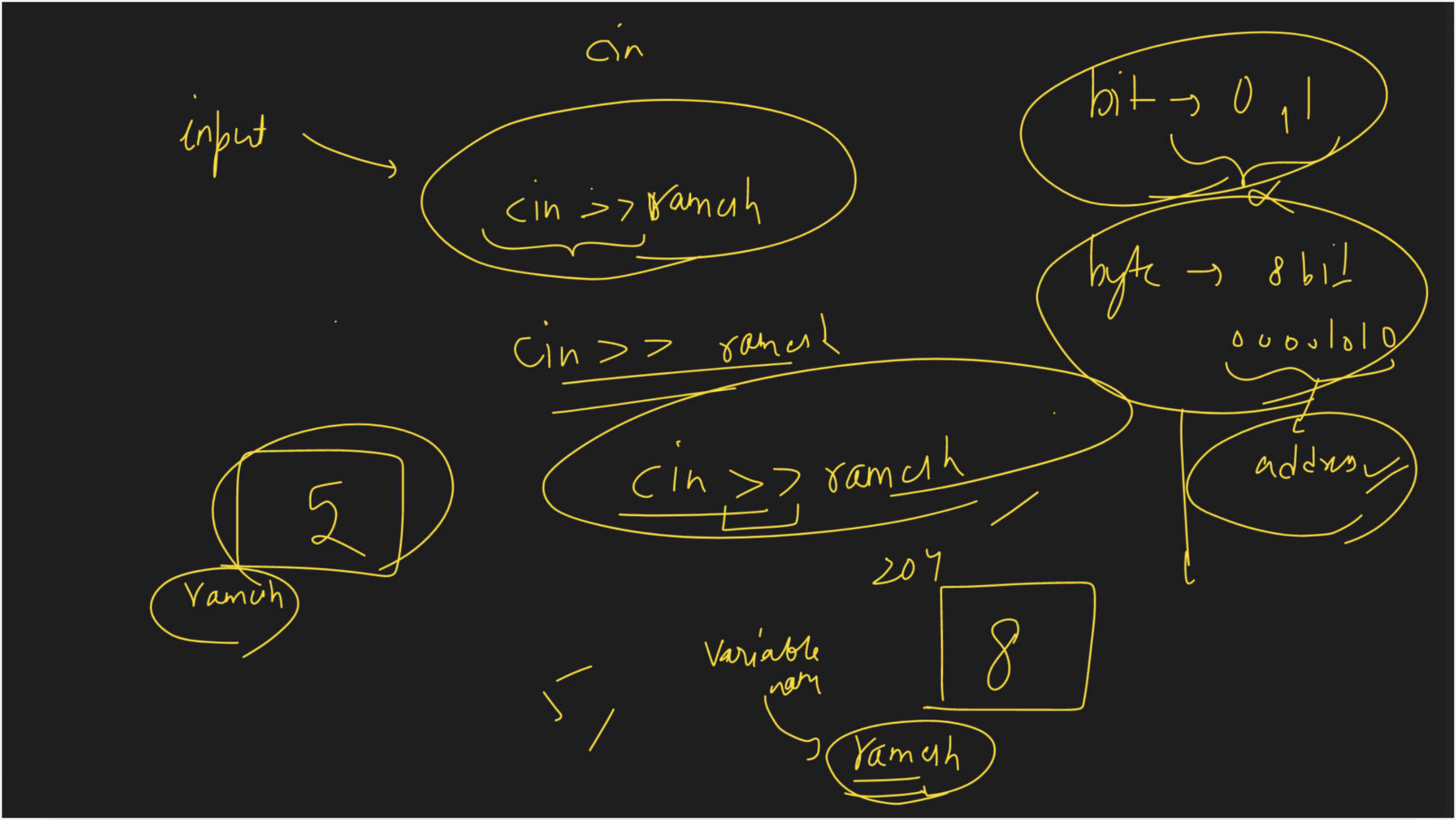

#include <iostream>
using namespace std;
int main()
cout << "Love";
return 0;
Syntax
rule
namespace
std
std
cout
cout
cout


iostream
namespace
find iosh
andar → code
std → open
1 min
Papa ka
Naam
Run


cout << 5
cout << 2;
alternative
H/w → newline character
52
endl
52
5
2
endl
cout << 5 << endl
cout << endl << endl << 2 << endl;
5
2
2

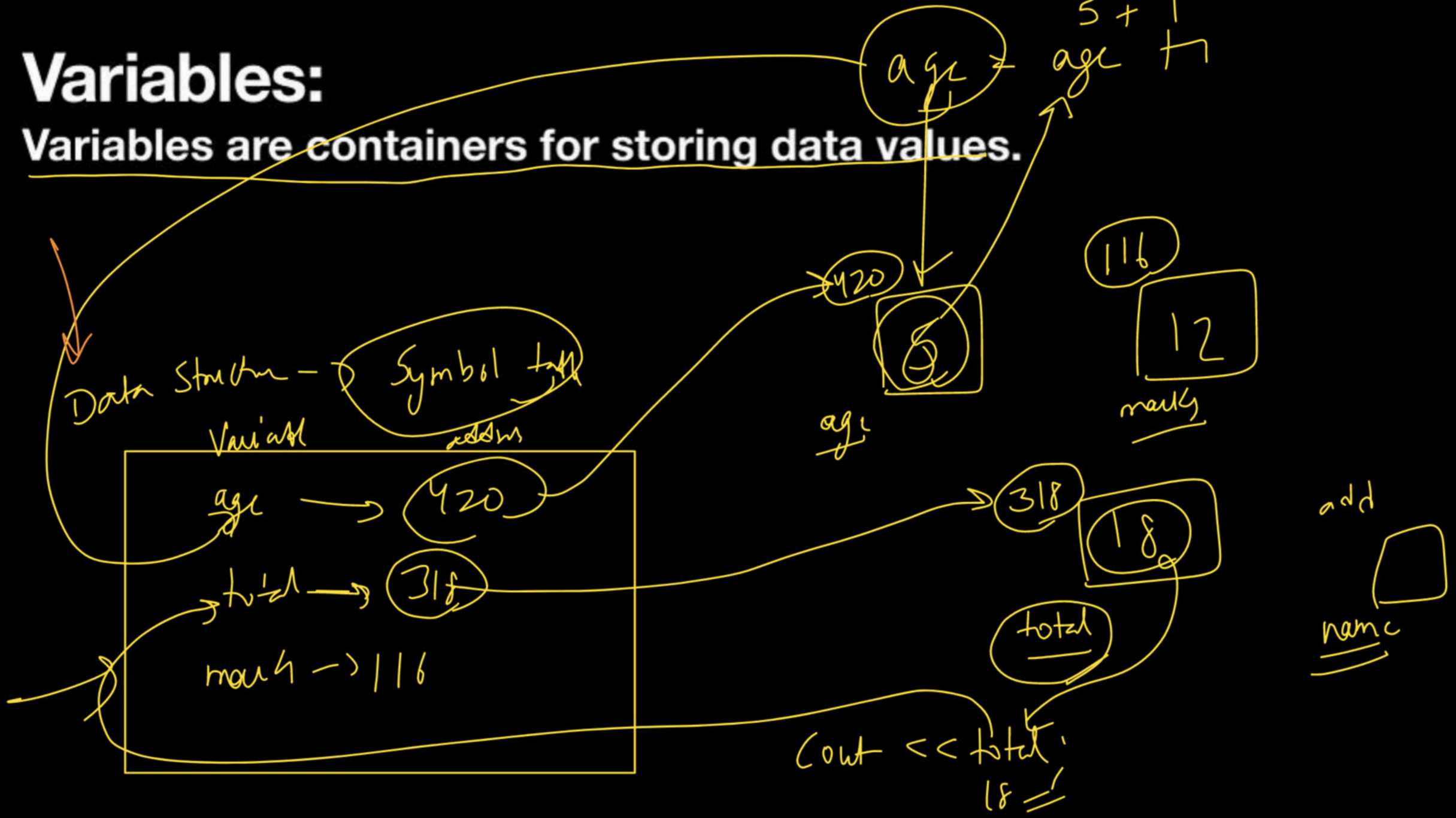
Input / Output in C++ :

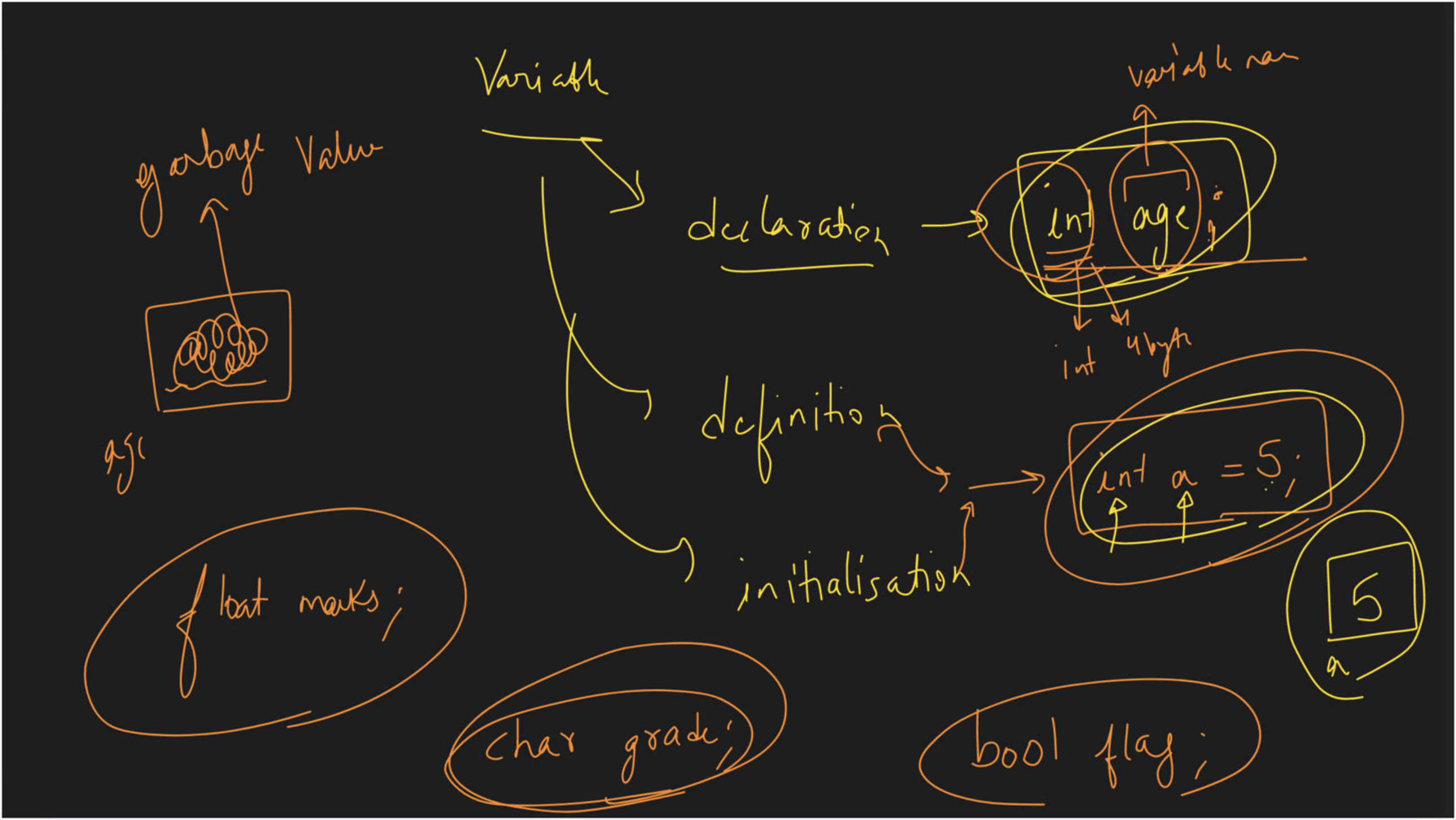


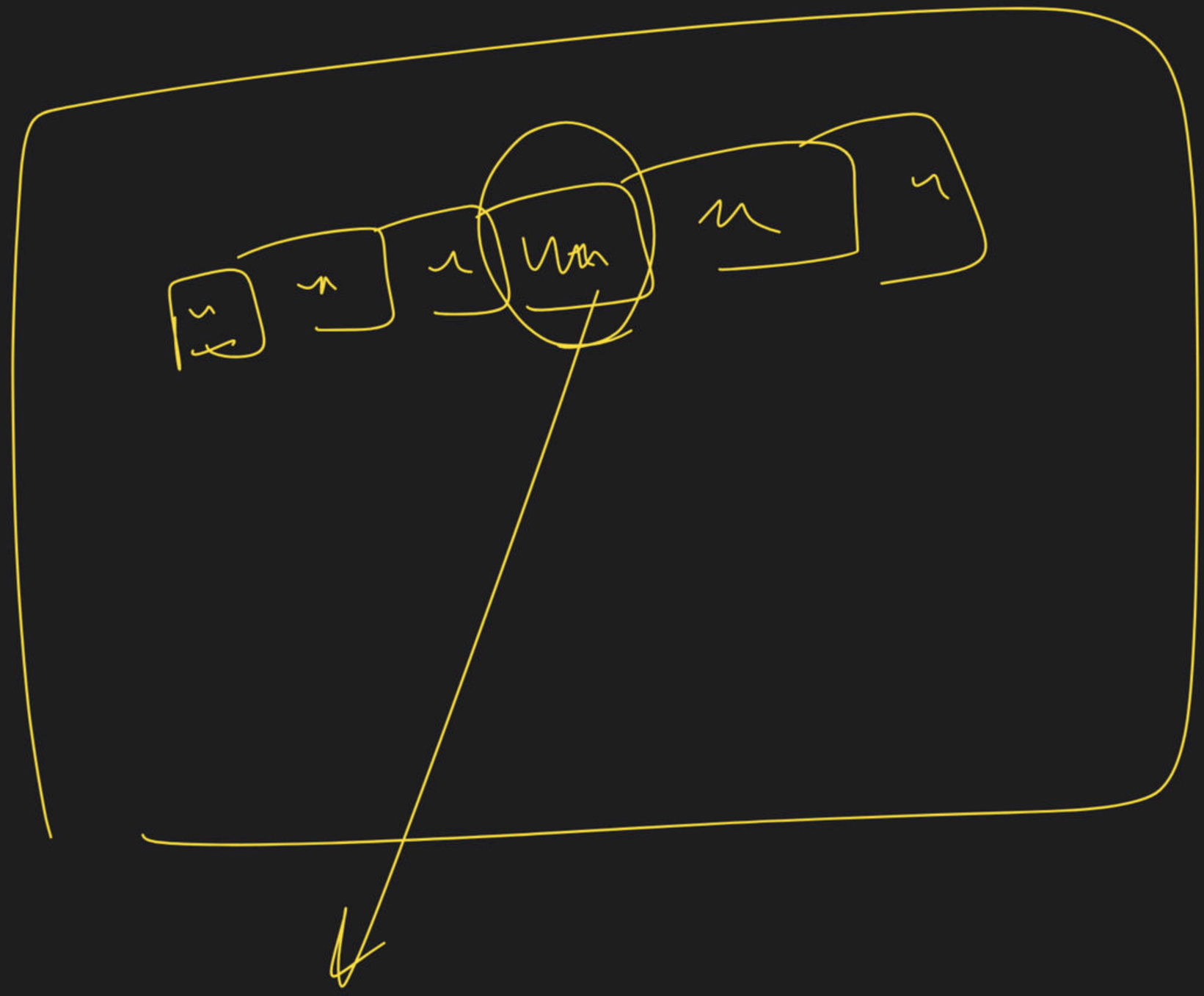

cin
input
cin >> ramesh
bit → 0 1
byte → 8 bit
00001010
address
cin >> ramesh
cin >> ramesh
5
ramesh
204
8
Variable name
ramesh

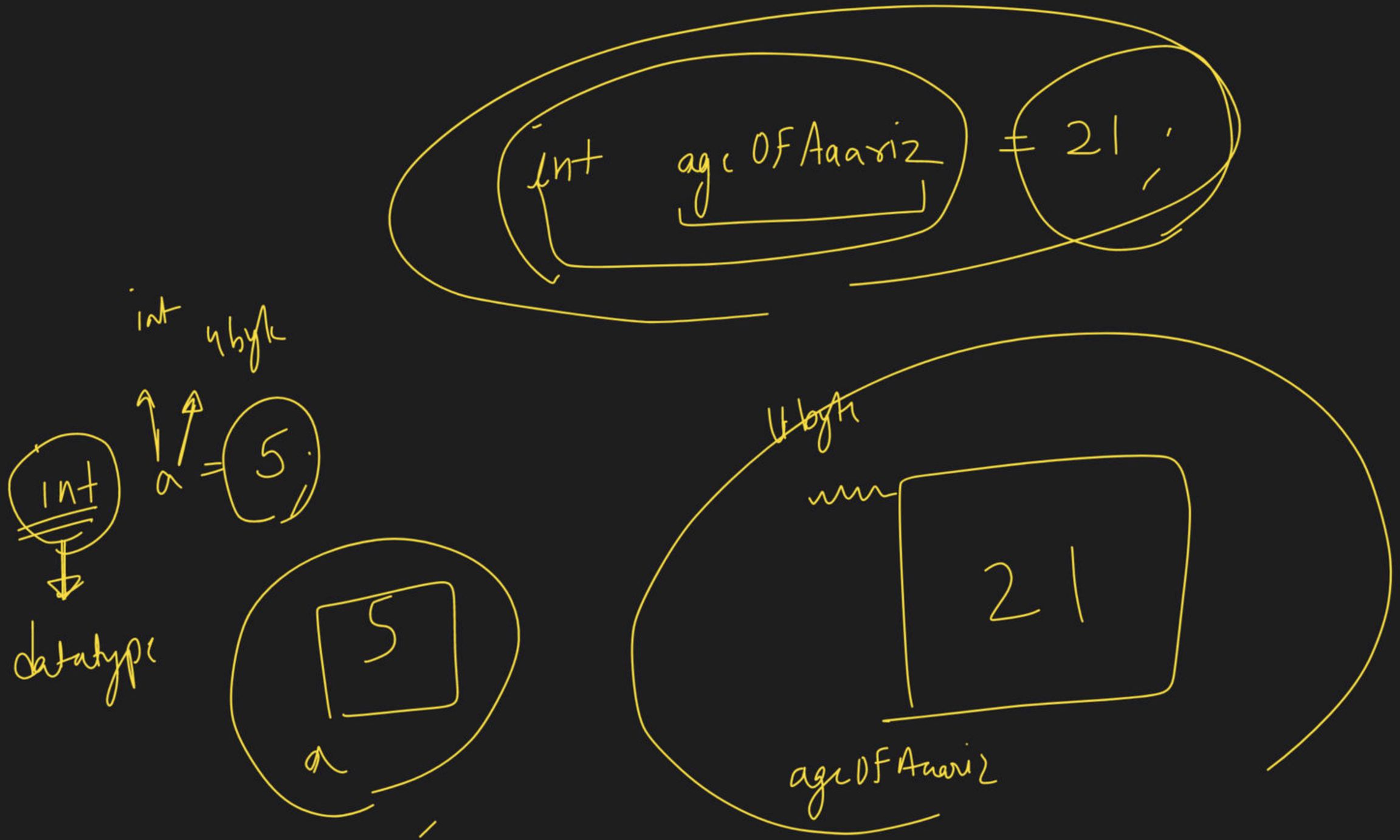
Variables:

Variables are containers for storing data values.




Variable
garbage Value
garbage
declaration
variable name
int age;
int
4byte
definition
int a = 5;
initialisation
5
a
float marks;
char grade;
bool flag;




int ageOfAaariz = 21;
int
4byte
a = 5;
int
datatype
5
a
4byte
21
ageOfAaariz

Datatypes

int → -ve true 0

char → 'a' 'b' 'c' ...

boolean → true / false

void → Bracdme

float → 1.23, 0.17, 3.14207/7215
double

```
bool babban = true;
            = 1;
```

true ↔ 1

false ↔ 0

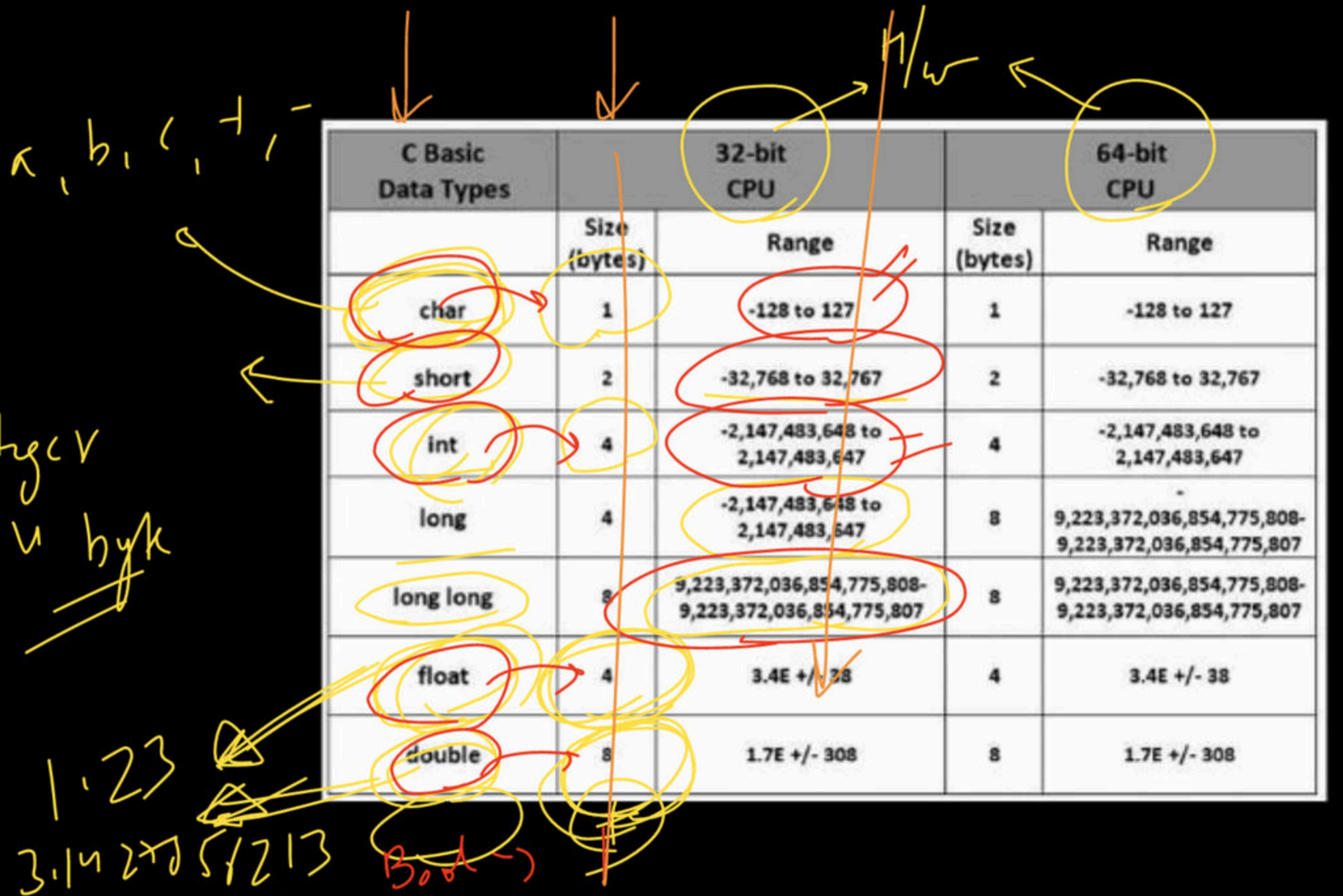
```
char grade = '+';
```

Datatypes:

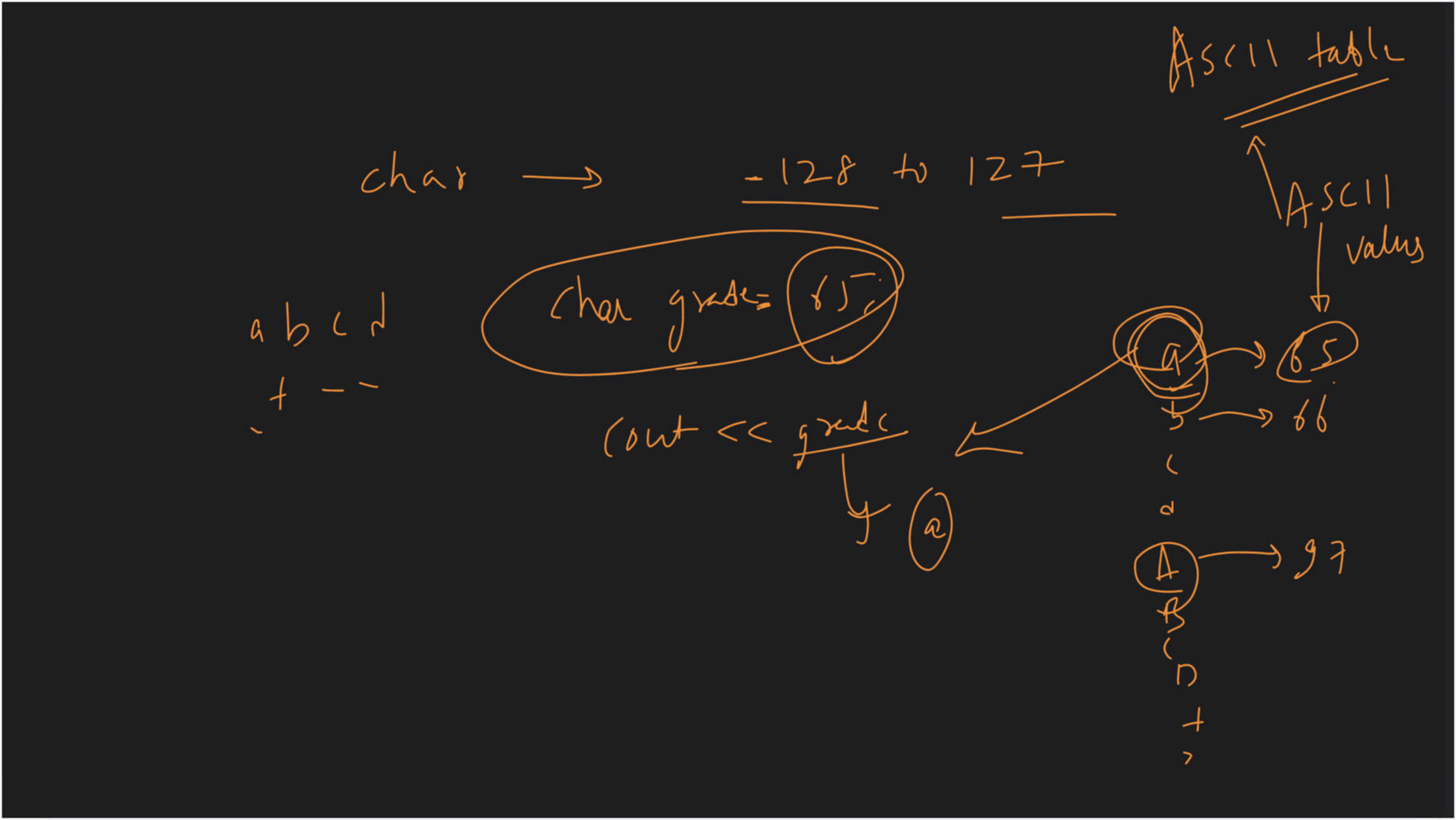
type of data → size of data

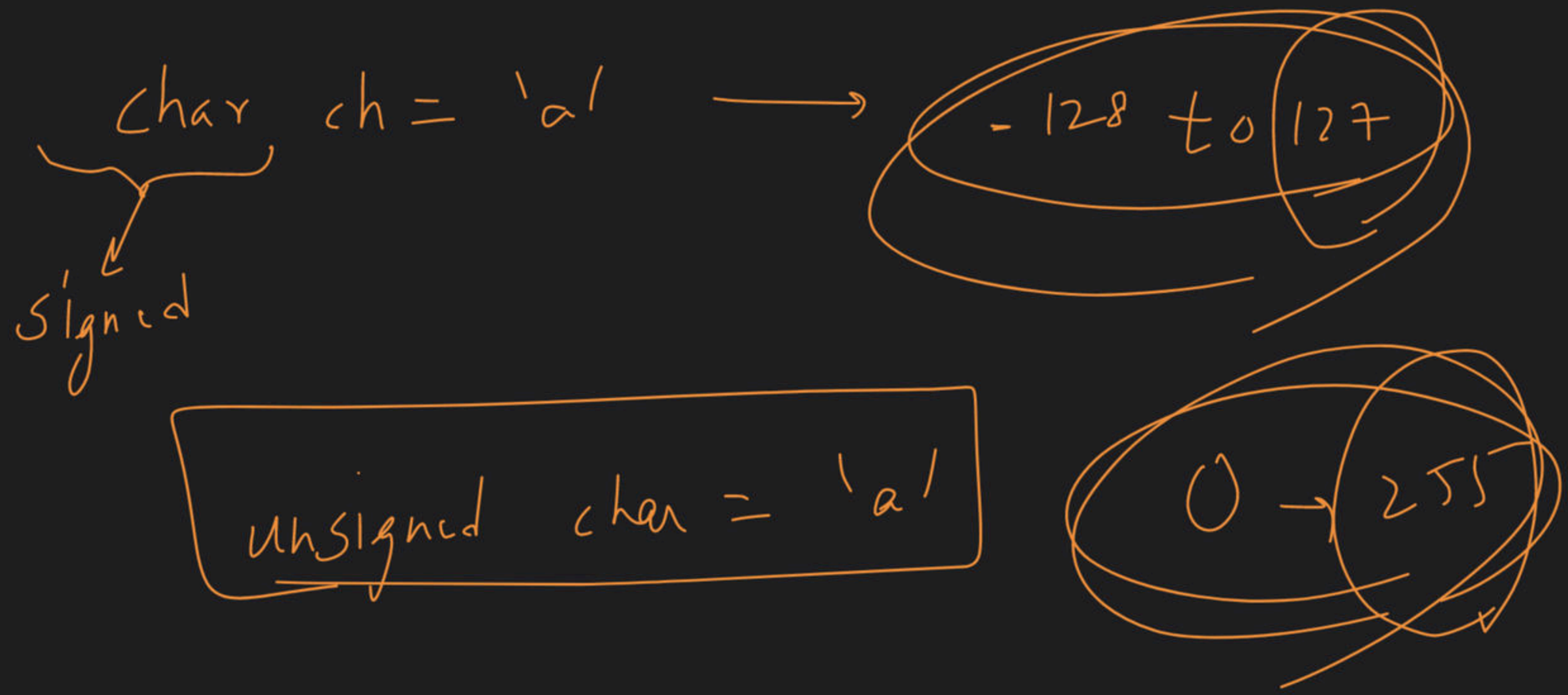
The data type specifies the size and type of information the variable will store

- Diagram



C Basic Data Types	32-bit CPU		64-bit CPU	
	Size (bytes)	Range	Size (bytes)	Range
char	1	-128 to 127	1	-128 to 127
short	2	-32,768 to 32,767	2	-32,768 to 32,767
int	4	-2,147,483,648 to 2,147,483,647	4	-2,147,483,648 to 2,147,483,647
long	4	-2,147,483,648 to 2,147,483,647	8	-9,223,372,036,854,775,808-9,223,372,036,854,775,807
long long	8	9,223,372,036,854,775,808-9,223,372,036,854,775,807	8	9,223,372,036,854,775,808-9,223,372,036,854,775,807
float	4	3.4E +/- 38	4	3.4E +/- 38
double	8	1.7E +/- 308	8	1.7E +/- 308


ASCII table
char → -128 to 127
ASCII values
char grade = 65;
a b c d
t - -
cout << grade;
y
@
A
65
66
a
97
B
C
D
t
?


char ch = 'a'
signed
-128 to 127
unsigned char = 'a'
0 → 255

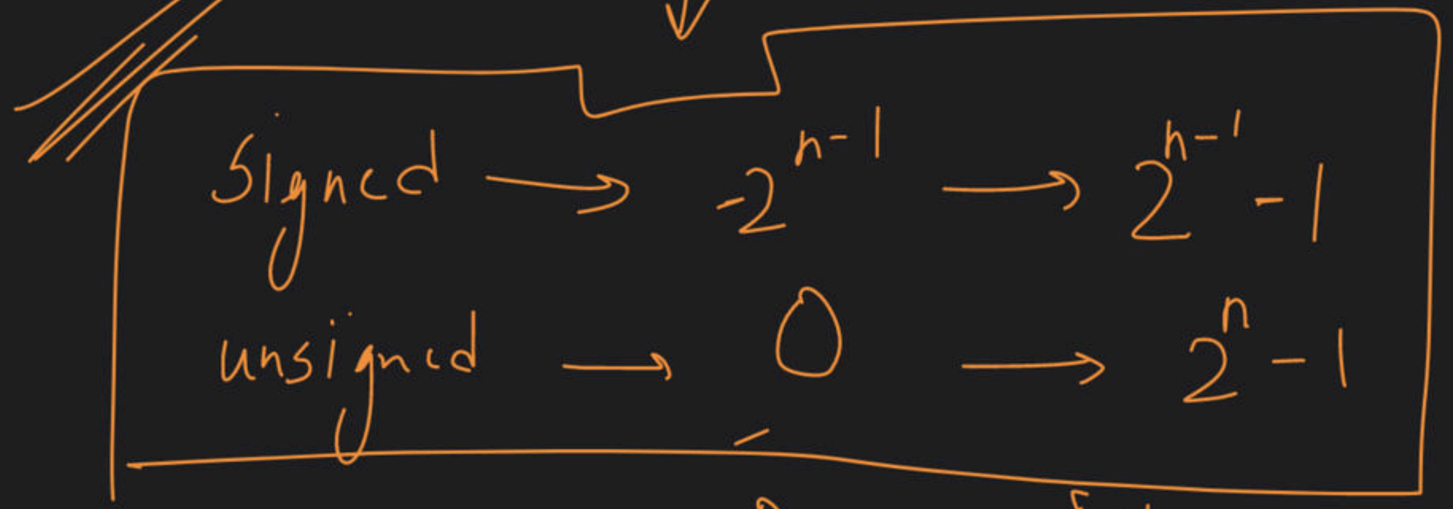
datatype

range

n = no of bits

char → 1byte → 8 bits

n = 8



Signed = $-2^{8-1} \rightarrow 2^{8-1}-1$

$-2^7 \rightarrow 2^7-1$

$-128 \rightarrow 128-1$

$-128 \rightarrow 127$

$0 \rightarrow 2^{8-1} \rightarrow 0 \rightarrow 255$

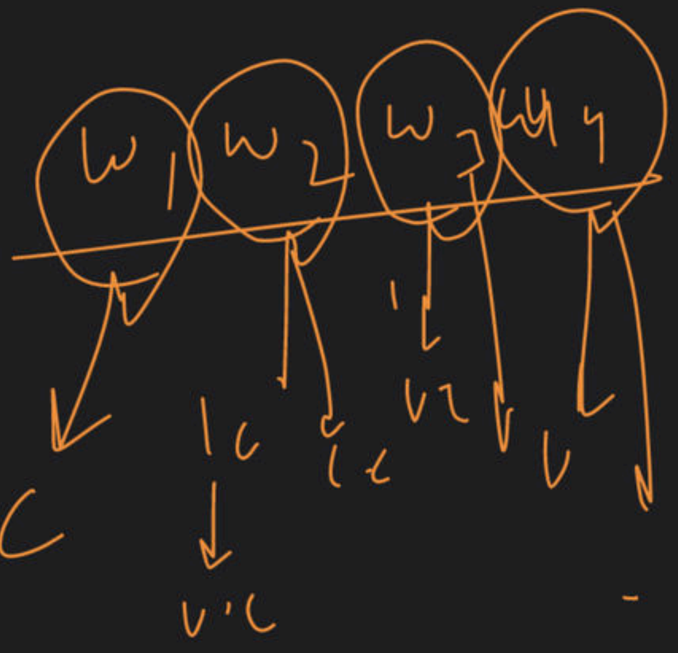
$0 \rightarrow 255$

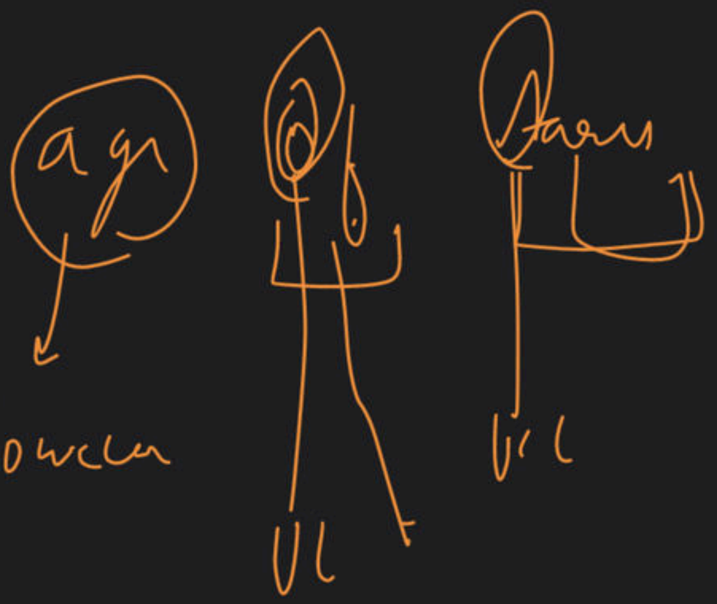
Variable Naming Conventions:

Naming Conventions rules for Variables are:

- ☑ It should begin with an alphabet.
- ☑ There may be more than one alphabet, but without any spaces between them.
- ☑ Digits may be used but only after alphabet.
- ☑ No special symbol can be used except the underscore (_) symbol. When multiple words are needed, an underscore should separate them.
- ☑ No keywords or command can be used as a variable name.
- ☑ All statements in C++ language are case sensitive. Thus a variable **A** (in uppercase) is considered different from a variable declared **a** (in lowercase).

```
int love=5;
int Love=5;
```





love babbar

age Of Aaariz



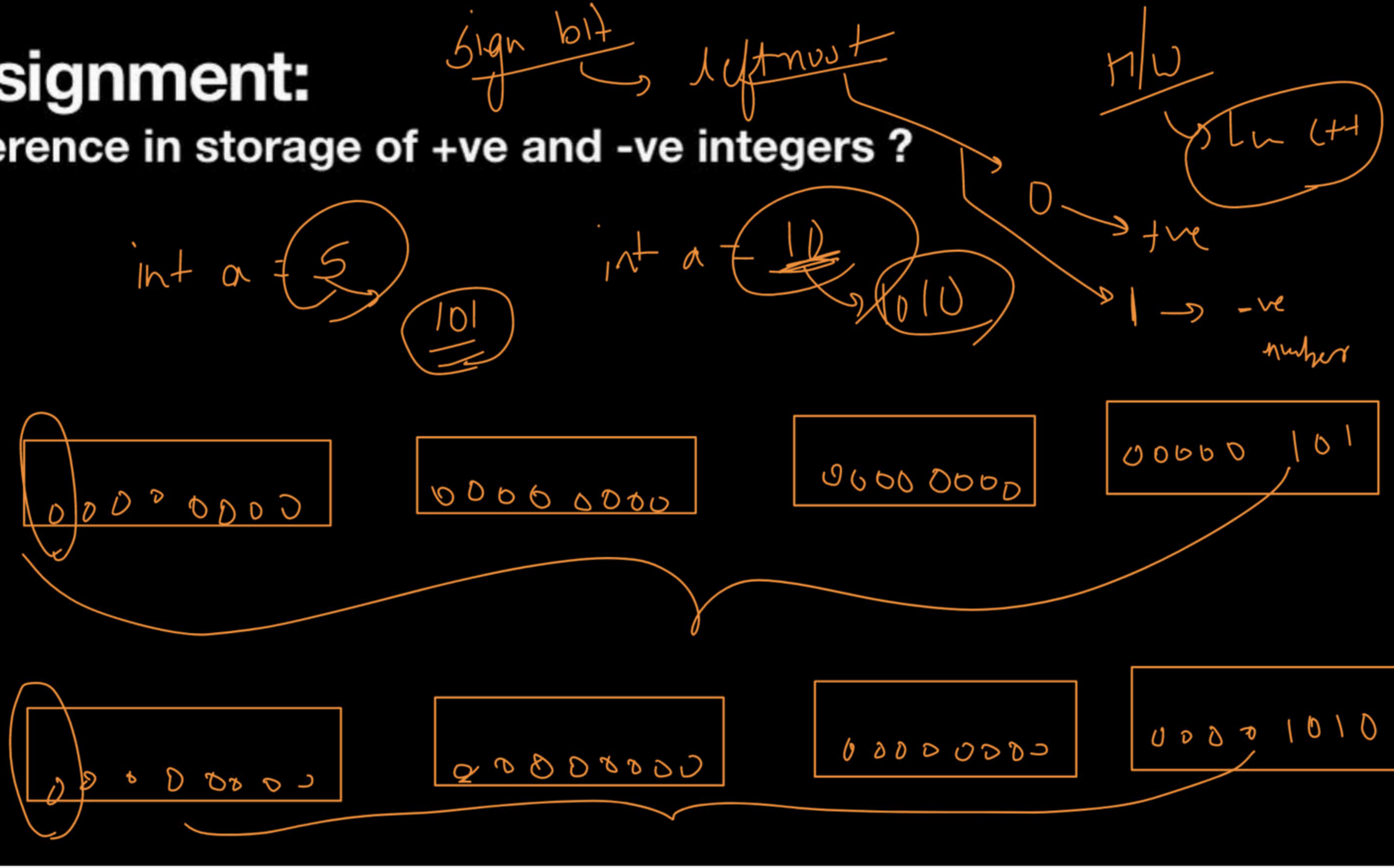
loveBabbar

harsh Kapoor

aditya Kumar

Assignment:

Difference in storage of +ve and -ve integers ?



1's complement

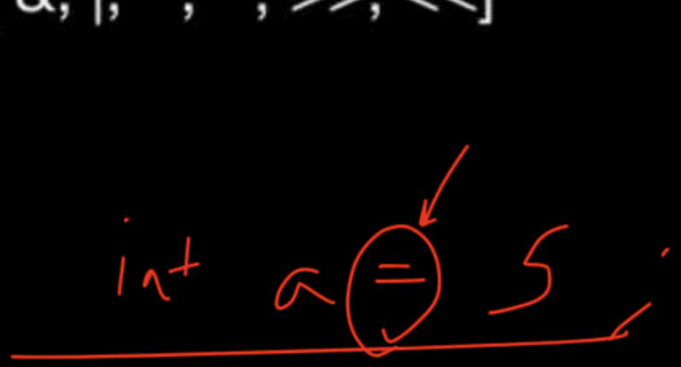
2's complement

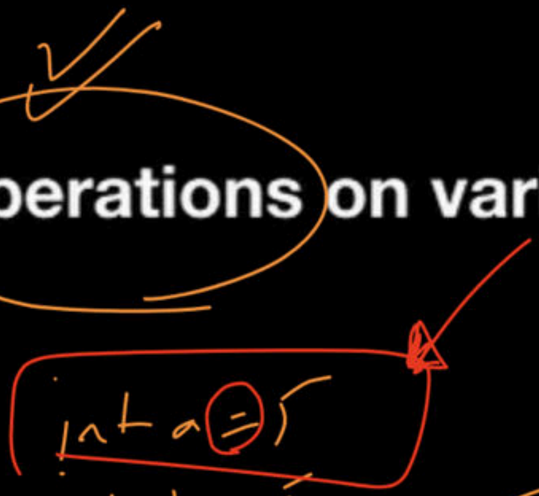
Signed vs Unsigned Integers:

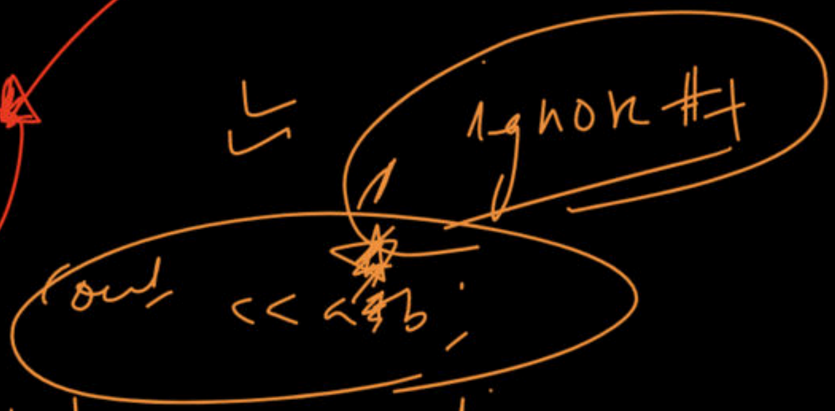
Operators:

Operators are used to perform operations on variables and values.

- Arithmetic [+, -, *, /, %, ++, - -]
- Relational [==, !=, >, <, >=, <=]
- Assignment [=]
- Logical [&&, ||, !]
- Bitwise [&, |, ~, ^, >>, <<]







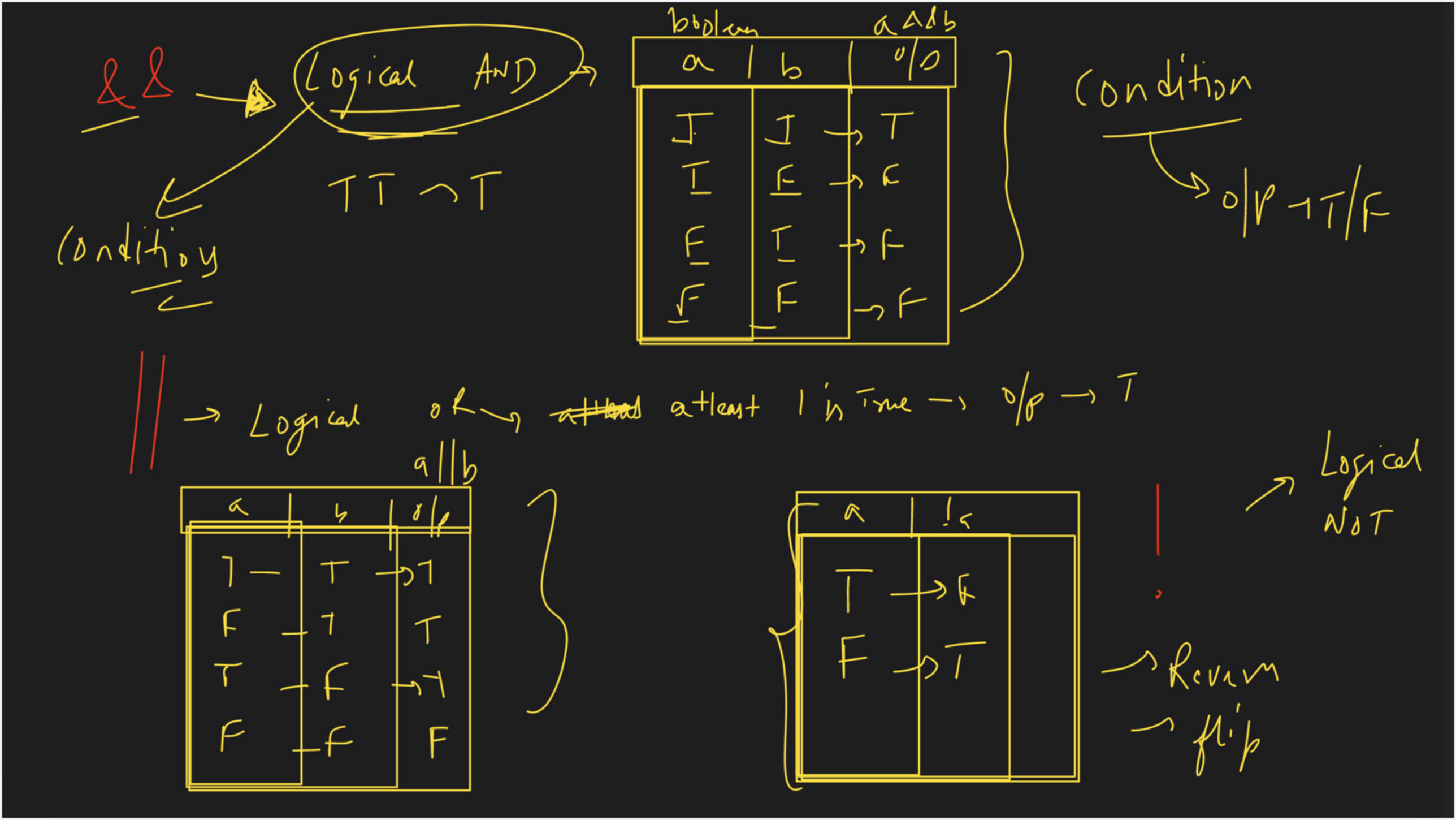
Symbol → operation

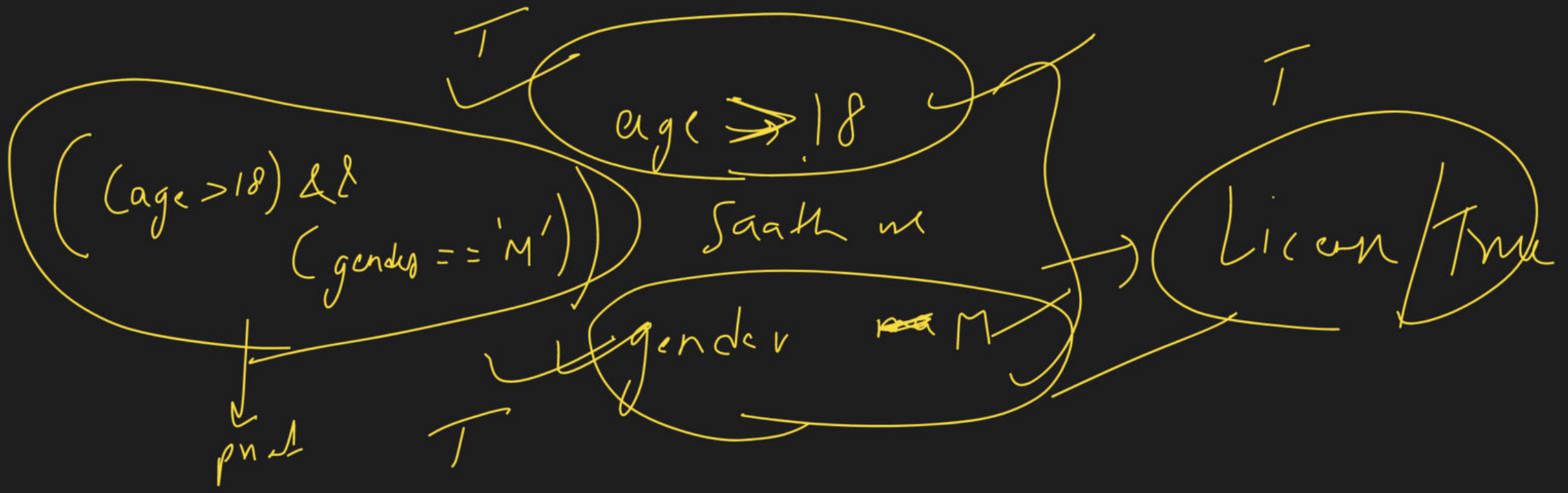
\+ → add

\- → sub

* → mult

/ → divid


&&
Logical AND
TT → T
Condition
bool
a && b
a | b | o/p
T | T → T
T | F → F
F | T → F
F | F → F
Condition
o/p → T/F
|| → Logical OR → atleast 1 is true → o/p → T
a || b
a | b | o/p
T | T → T
F | T → T
T | F → T
F | F → F
! → Logical NOT
a | !a
T → F
F → T
Return flip

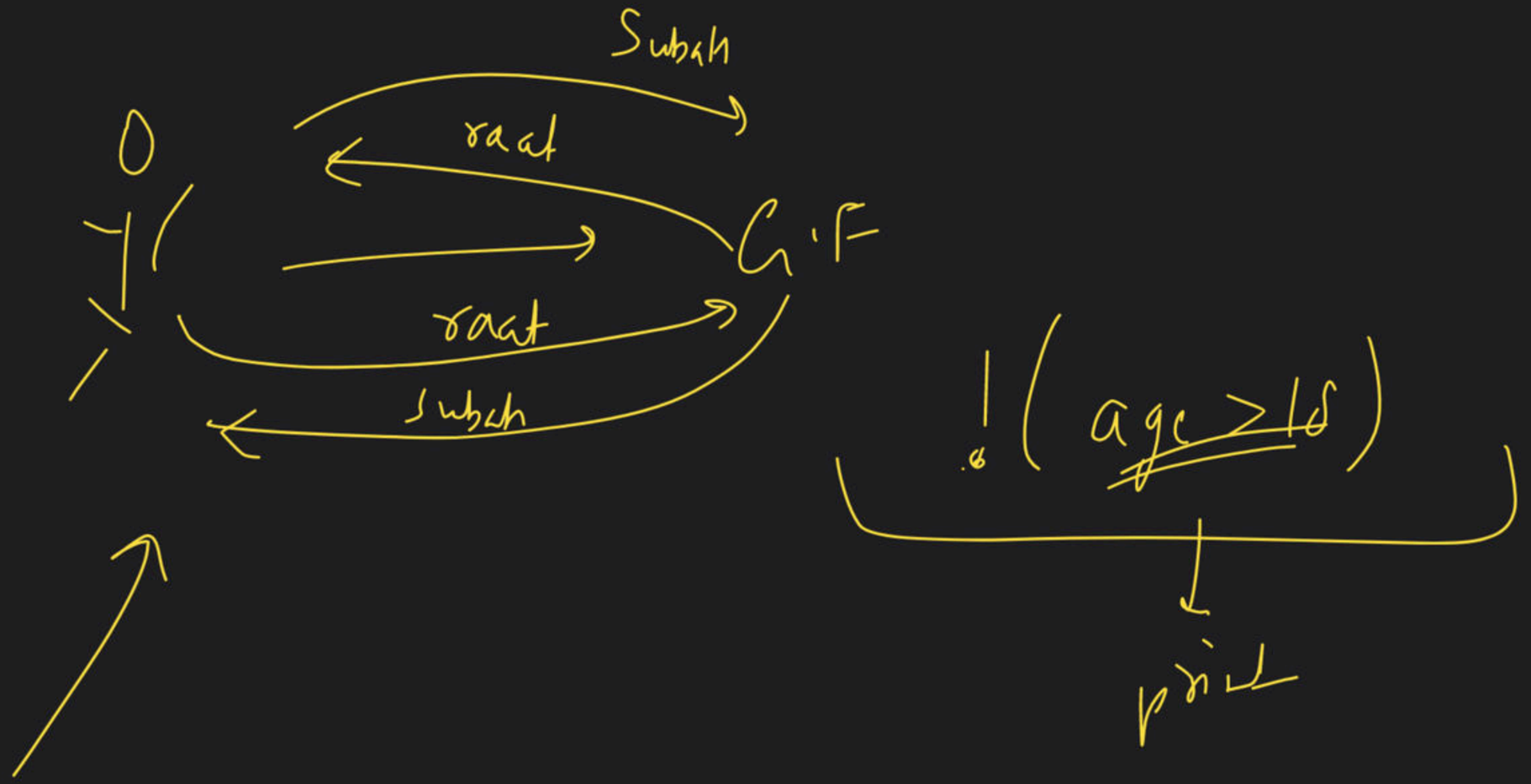


else {

return False

}

((age > 18) || (gender == 'M'))


Subah
raat
raat
subah
!(age>18)
pani

Assignment:

Complete week-“Learn C++” recorded videos.

- 32 bit vs 64 bit Architecture
- Typecasting: Implicit and Explicit
- Binary and Decimal Number System

